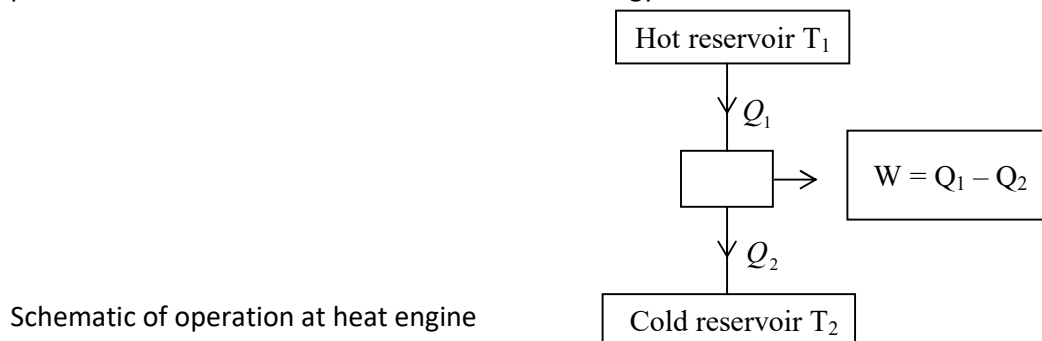


Second Law of Thermodynamics

2. Heat Engines

A machine that can convert heat into work is said to be heat engine. It is a system that performs the conservation of heat or thermal energy to mechanical work.



Heat Reservoir

It is an effectively infinite pool of thermal energy of a given, constant temperature. Ideal its heat capacity is large enough that when it is in thermal contact with another system, its temperature remains constant.

All heat engines have mainly three essential components.

1. A source: This is a hot region which is a part of the surrounding from which energy flows by heat transfer. Popularly it is known as hot reservoir. Example is a nuclear reactor, furnace.
2. The working agent: This under goes change at state as part of a continuous cycle e.g. steam-water.
3. A sink: This is a respectively cold region which is part or the surrounding into which heat is rejected by heat transfer e.g. cooler.

Efficiency of Heat Engine (η):

$$\eta = \frac{\text{work done by the system}}{\text{heat given into the system}}$$

Hence the process is cyclic $dU = 0$.

$$dQ = dW$$

Heat given to system is Q_1 and heat rejected by system is Q_2 .

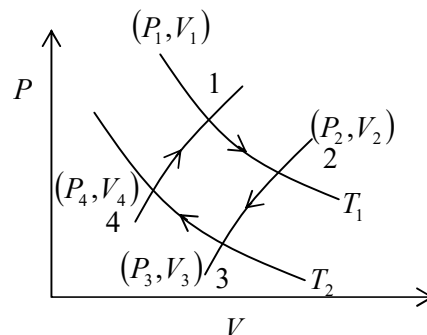
$$\eta = \frac{Q_1 - Q_2}{Q_1}$$

Carnot Cycle: It is theoretical thermodynamics cycle proposed by Nicolas, Leonard Sadi Carnot.

It can be shown that it is most efficient cycle for converting a given amount of thermal energy into work.

A system undergoing a Carnot cycle is called a Carnot heat engine.

Stages of the Carnot cycle:



1. Reversible isothermal expansion of the gas at hot temperature, T_1 (isothermal heat addition). During this step (1 to 2) the gas is allowed to expand and it does work on the surrounding. The gas expansion is propelled by absorption of quantity Q_1 at heat from the high temperature T_1 .
2. Reversible adiabatic expansion of the gas. For this step (2 to 3) on figure, the gas continues to expand, working on surrounding. The gas expansion causes it to cool to the cold temperature T_2 .
3. Reversible isothermal compression of the gas at the “cold” temperature T_2 (isothermal heat rejection). The process is shown by $3 \rightarrow 4$. Now the surroundings do work on the gas, causing quantity Q_2 of heat to flow out of the gas to the low temperature reservoir.
4. Adiabatic compression of the gas (4 to 1). During this step, the surrounding do work on the gas, compressing it and cause the temperature to rise to T_1 . At this point the gas in the same state as at start at step 1.

Efficiency of Carnot engine:

$$\eta = \frac{W}{Q_1}$$

W is the work done during cycle and Q_1 is heat given to system.

$$W = W_1 + W_2 + W_3 + W_4$$

$$W_1 = \text{work done during process 1 to 2 for isothermal process} = nRT_1 \ln \frac{V_2}{V_1}$$

change in internal energy during the process $dU = 0$

$$Q_1 = W_1 = nRT_1 \ln \left(\frac{V_2}{V_1} \right) \text{ (From first law of thermodynamics)}$$

W_2 is work done during process 2 to 3 in adiabatic process.

$$dQ = 0 \quad W_2 = \frac{nR(T_2 - T_1)}{1 - \gamma}$$

W_3 = work done during process 3 to 4 isothermal compression.

$$W_3 = nRT_2 \ln \left(\frac{V_4}{V_3} \right)$$

$$Q_2 = W_3 \quad dU = 0 \quad (\text{isothermal process})$$

W_4 , the work done during the adiabatic process 4 to 1, is given by

$$W_4 = \frac{nR(T_1 - T_2)}{1 - \gamma}$$

$$W_2 = -W_4$$

$$W = nRT_1 \ln\left(\frac{V_2}{V_1}\right) + nRT_2 \ln\left(\frac{V_4}{V_3}\right)$$

$$Q_1 = nRT_1 \ln\left(\frac{V_2}{V_1}\right)$$

$$\eta = \frac{W}{Q_1} = 1 + \frac{T_2}{T_1} \frac{\ln\left(\frac{V_4}{V_3}\right)}{\ln\left(\frac{V_2}{V_1}\right)} \quad \dots(i)$$

From the figure,

$$T_1 V_2^{\gamma-1} = T_2 V_3^{\gamma-1} \quad \dots(ii)$$

$$T_2 V_4^{\gamma-1} = T_1 V_1^{\gamma-1} \quad \dots(iii)$$

Dividing equation (ii) by (iii) gives

$$\left(\frac{V_2}{V_1}\right)^{\gamma-1} = \left(\frac{V_3}{V_4}\right)^{\gamma-1}$$

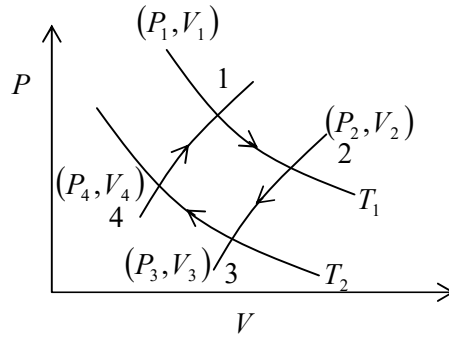
$$\Rightarrow \frac{V_2}{V_1} = \frac{V_3}{V_4} \quad \dots(iv)$$

From equations (i) and (iv)

$$\eta = 1 - \frac{T_2}{T_1} \quad \dots(v)$$

For Carnot cycle, $\frac{Q_2}{Q_1} = \frac{T_2}{T_1}$

This gives $\eta = 1 - \frac{Q_2}{Q_1} \quad \dots(vi)$



Example: An ideal gas engine operator in a cycle which when represent on a P - V diagram is a rectangle. If we call P_1, P_2 and the lower and higher pressures respectively and V_1, V_2 as lower and higher volume respectively.

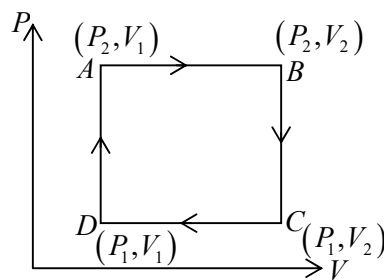
- Calculate the work done in complete cycle
- indicate in which parts of the cycle heat is absorbed and in which part librated.
- Calculate the quantity of heat following into the gas in one cycle
- show that efficiency of the engine is

$$\eta = \frac{\gamma - 1}{\frac{\gamma P_2}{P_2 - P_1} + \frac{V_1}{V_2 - V_1}}$$

Solution: work done drawing $AB = P_2(V_2 - V_1)$

Heat absorbed = $nC_p \Delta T$

$$\begin{aligned} &= nR \frac{\gamma}{\gamma - 1} (T_2 - T_1) = \frac{\gamma}{\gamma - 1} (P_2 V_2 - P_2 V_1) \\ &= \frac{\gamma}{\gamma - 1} P_2 (V_2 - V_1) \end{aligned}$$



$V_2 > V_1$ so heat absorbed in the process.

In process B-C isochoric process

$$W_{B-C} = 0$$

$$dQ = dU$$

$$nC_v \Delta T$$

$$\frac{n}{\gamma - 1} R \Delta T = \frac{V_2}{\gamma - 1} (P_1 - P_2)$$

$$P_1 < P_2 \text{ heat rejected}$$

In process C-D isobaric process

$$W_{C-D} = P_1(V_1 - V_2)$$

Heat exchange during the process

$$Q_{C \rightarrow D} = \frac{\gamma}{\gamma - 1} P_1 (V_1 - V_2)$$

$V_1 < V_2$ so heat is rejected

In process D $\rightarrow$ A isochoric process

$$W_{D \rightarrow A} = 0$$

Heat exchange during the process

$$nC_v \Delta T = \frac{nR \Delta T}{\gamma - 1} = \frac{V_1 (P_2 - P_1)}{\gamma - 1} \text{ heat absorbed.}$$

(a) Work done = $P_2 (V_2 - V_1) + P_1 (V_1 - V_2)$

$$W = (P_2 - P_1) (V_2 - V_1)$$

(b) Heat absorbed $\frac{\gamma}{\gamma - 1} P_2 (V_2 - V_1) + \frac{V_1}{\gamma - 1} (P_2 - P_1)$

(c) Heat flowing into the cycle

$$\frac{\gamma P_2 (V_2 - V_1)}{\gamma - 1} + \frac{V_2}{\gamma - 1} (P_1 - P_2) + \frac{\gamma}{\gamma - 1} P_1 (V_1 - V_2) + \frac{\gamma}{\gamma - 1} V_1 (P_2 - P_1)$$

$$= \frac{\gamma}{\gamma - 1} (V_2 - V_1) (P_2 - P_1) + \frac{1}{\gamma - 1} (P_1 - P_2) \cdot (V_2 - V_1)$$

$$= \frac{\gamma}{\gamma - 1} (V_2 - V_1) (P_2 - P_1) - \frac{1}{\gamma - 1} (P_2 - P_1) (V_2 - V_1)$$

$$(P_2 - P_1) (V_2 - V_1) \left[\frac{\gamma}{\gamma - 1} - \frac{1}{\gamma - 1} \right] = (P_2 - P_1) (V_2 - V_1)$$

$$dQ = dW$$

(d) Efficiency = $\frac{W}{Q_{\text{absorbed}}} = \frac{(P_2 - P_1) (V_2 - V_1)}{\frac{\gamma P_2 (V_2 - V_1)}{\gamma - 1} + \frac{V_1 (P_2 - P_1)}{\gamma - 1}}$

Therefore, $\eta = \frac{\gamma - 1}{\gamma \frac{P_2}{P_2 - P_1} + \frac{V_1}{V_2 - V_1}}$